

Power BI - Financial Data Training Guide

Lockheed Martin

Overview

This training uses sample financial data modeled after Dynamics 365 Finance & Operations and Dataverse table structures. Participants will learn to connect Power BI to Dataverse, build financial reports, and publish them securely within a GCC High environment.

Prerequisites

- Power BI Desktop (GCC High version) installed
 - Access to a Dynamics 365 Finance & Operations environment (or Dataverse sandbox)
 - Appropriate security roles: **Finance Manager** or **Financial Analyst** in Dynamics 365
 - Power BI Pro or Premium Per User license (GCC High tenant)
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Sample Data Files

The following CSV files are provided in the **Training Data/** folder for hands-on exercises:

File	Description
Financial Data - Programs.csv	Program dimension table (lookup/reference)
Financial Data - Budget vs Actuals.csv	Program-level budget tracking with variance analysis
Financial Data - Purchase Orders.csv	Procurement data with vendor and contract details

Connecting to Dataverse from Power BI

Method 1: Direct Dataverse Connector

1. Open Power BI Desktop
2. Select **Get Data > Dataverse**
3. Enter your environment URL: <https://yourorg.crm9.dynamics.com> (GCC High uses **.crm9**)
4. Authenticate with your organizational credentials
5. Select the required tables from the Navigator pane
6. Click **Transform Data** to open Power Query Editor

Method 2: Dynamics 365 OData Feed

1. Select **Get Data > OData Feed**
2. Enter: <https://yourorg.crm9.dynamics.com/api/data/v9.2>
3. Select **Organizational Account** for authentication

4. Choose relevant entities

Method 3: Import from CSV (Training Only)

- 1. Select **Get Data** > **Text/CSV**
- 2. Navigate to the **Training Data/** folder
- 3. Select the financial data CSV files
- 4. Review column types in the preview and click **Load**

Data Model Relationships (Star Schema)

Set up the following relationships in Power BI Model View:

```
Programs[Program ID] (1) → Budget vs Actuals[Program ID] (Many)
Programs[Program ID] (1) → Purchase Orders[Program ID] (Many)
```

The **Programs** table acts as a dimension/lookup table with one row per program. Both fact tables (Budget vs Actuals and Purchase Orders) connect to it, enabling cross-filtering between budget data and procurement data through the shared dimension.

Using Aggregations in Visuals (No Formulas Needed)

Power BI automatically calculates totals, averages, and counts when you drag fields into visuals. You do NOT need to write any formulas.

How to change the aggregation on a field:

- 1. Drag a number field (e.g., **Budgeted Amount**) into a visual
- 2. Click the dropdown arrow next to the field name in the **Values** area
- 3. Choose: Sum, Average, Count, Min, Max, etc.

Common aggregations for this training data:

Visual Need	Field to Use	Aggregation
Total Budget	Budgeted Amount	Sum
Total Actual Spend	Actual Amount	Sum
Total PO Value	PO Amount	Sum
Number of POs	PO ID	Count
Total Outstanding POs	Amount Remaining	Sum

Tip: If Power BI shows "Count" when you expected "Sum", it means the column is still typed as Text. Go back to Power Query and change the data type to Decimal Number.

Exercises

Exercise 1: Load and Transform Data

1. Import all 3 CSV files into Power BI Desktop
2. In Power Query, clean the data following POWER_QUERY_TRANSFORMATIONS.md
3. Set appropriate data types (Currency, Percentage, Date)
4. Establish relationships between tables in Model View (Programs → Budget vs Actuals, Programs → Purchase Orders)

Exercise 2: Build a Budget vs Actuals Dashboard

1. Create a clustered bar chart comparing Budget vs Actuals by Program
2. Add a KPI card showing total variance
3. Add conditional formatting (red for over-budget, green for on-track)
4. Add slicers for Fiscal Year and Quarter

Exercise 3: Cross-Data Executive Summary

1. Create a Matrix with Programs as rows, Quarters as columns, showing Actual Amount
2. Add KPI Cards: Total Budget, Total Actual, Total PO Value, Total Outstanding
3. Build a line chart showing spending trend by Program across FY2025–FY2026
4. Add a Fiscal Year slicer to compare year-over-year performance

Exercise 4: Publish to Power BI Service

1. Publish the report to a Power BI GCC High workspace
2. Set up a scheduled refresh connecting to Dataverse

Troubleshooting

Issue	Solution
"Cannot connect to Dataverse"	Verify GCC High URL uses .crm9.dynamics.com
Tables not visible	Check Dataverse security role has Read access
Refresh fails in Service	Configure gateway for on-premises data; use service principal for Dataverse
Slow query performance	Use Import mode instead of DirectQuery for large datasets
Missing financial dimensions	Map D365 financial dimension sets in Power Query

Additional Resources

- [Power BI for US Government Documentation](#)
- [Dataverse Connector for Power BI](#)
- [Dynamics 365 Finance Data Entities](#)
- [GCC High Environment URLs](#)

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